

RETAIL SALES ANALYSIS

SQL Business Analysis | PostgreSQL

₹458,895
2023 Revenue

₹365,141.70
2023 Profit

2.13%
Revenue Growth

3.10%
Profit Growth

Project Objective

Analyze retail transactions with PostgreSQL to understand revenue, profitability, customer behavior, sales trends, time-of-day performance and category contribution, then translate the results into business insights.

Key Findings

- **Category performance:** Electronics generated the highest revenue; Clothing generated the highest total profit and led transactions, units sold and unique customers; Beauty had the highest profit margin and AOV.
- **Customer behavior:** Female customers had the highest revenue and AOV. The 18–25 group had the highest AOV, while the 40–59 group generated the highest revenue.
- **Customer concentration:** The top 10 customers generated ₹214,400, equal to 23.61% of total revenue. All 155 customers recorded in the dataset made at least two purchases.
- **Growth:** Revenue increased 2.13% from 2022 to 2023, while profit increased 3.10%, meaning profit grew faster than revenue.
- **Monthly trend:** December was the highest-revenue and highest-profit month in both years. February 2022 and March 2023 were the lowest-revenue months.
- **Time of day:** Evening led transactions, units sold, revenue and profit. Morning had the highest AOV and profit margin.
- **Category mix:** Revenue was balanced across Electronics (34.29%), Clothing (34.13%) and Beauty (31.58%), indicating low dependence on a single category.

Business Interpretation

The analysis shows that revenue, profit, demand and margin do not tell the same story. Electronics leads revenue, Clothing is strongest on total profit and demand, and Beauty is strongest on margin. The business also shows a consistent December peak, strong evening volume and meaningful revenue concentration among top customers.

Recommendations

- **Customer retention:** Prioritize loyalty and retention efforts for high-value customers.
- **Seasonal planning:** Use the recurring December peak to plan inventory, staffing and promotions.
- **Category learning:** Investigate Clothing's profitability and Beauty's high margin for transferable pricing or cost practices.
- **Time-of-day planning:** Optimize Evening for volume and explore Morning for higher-value or margin-focused offers.

SQL Skills Demonstrated

Data cleaning checks • aggregation and GROUP BY • CASE • date/time functions • AOV and profit-margin calculations • subqueries • CTEs • LAG and RANK window functions • customer segmentation • business interpretation

Project focus: turning transaction-level SQL analysis into practical business insights.